

CO4 – AT2: Industry Based Problem Solving

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COURSE: FUNDAMENTAL DATA SCIENCE

COURSE CODE: DSA0405

Question 16 — Cybersecurity

A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate using a p-value.

Step-by-step solution

1. Population	All malicious activities (attacks) that the detection system may encounter.
2. Sample	200 attacks tested against the new detection algorithm.
3. Parameter of interest	Population proportion of malicious activities correctly detected p .
4. Point estimate	$\hat{p} = 194 / 200 = 0.97$ (97%).
5. Null hypothesis H_0	$p = 0.95$ (the new algorithm has NOT improved detection).
6. Alternative hypothesis H_1	$p > 0.95$ (the new algorithm HAS improved detection) — right-tailed test.

7. Significance level α	$\alpha = 0.05.$
8. Test statistic	$SE = \sqrt{[p_0(1 - p_0) / n]} = \sqrt{[(0.95 \times 0.05) / 200]} = \sqrt{0.0002375} = 0.01541$ $z = (\hat{p} - p_0) / SE = (0.97 - 0.95) / 0.01541 = 1.30$
9. p-value	$p\text{-value} = P(Z > 1.30) = 1 - 0.9032 = 0.0968.$
10. Compare p-value with α	$0.0968 > 0.05 \rightarrow p\text{-value is greater than } \alpha.$

Industry-oriented conclusion

Since the p-value (0.0968) exceeds α (0.05), we fail to reject H_0 . Although the sample detection rate rose numerically from 95% to 97%, there is not enough statistical evidence at the 5% significance level to conclude that the new algorithm has significantly improved the detection rate. From an industry standpoint, the cybersecurity team should continue monitoring performance over a larger sample before crediting the new algorithm with a genuine improvement.

Question 17 — Healthcare

A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with standard deviation 12 minutes. Test whether the new system significantly reduces waiting time using the p-value approach.

Step-by-step solution

1. Population	All patients visiting the hospital under the new scheduling system.
2. Sample	64 patients whose waiting times were recorded after implementation.
3. Parameter of interest	Population mean patient waiting time, μ .
4. Point estimate	$\bar{x} = 38$ minutes (sample standard deviation $s = 12$ minutes).
5. Null hypothesis H_0	$\mu = 42$ minutes (the new system has NOT reduced waiting time).
6. Alternative hypothesis H_1	$\mu < 42$ minutes (the new system HAS reduced waiting time) — left-tailed test.
7. Significance level α	$\alpha = 0.05$.
8. Test statistic	$SE = s / \sqrt{n} = 12 / \sqrt{64} = 12 / 8 = 1.5$

	$z = (\bar{x} - \mu_0) / SE = (38 - 42) / 1.5 = -2.67$
9. p-value	p-value = $P(Z < -2.67) \approx 0.0038$.
10. Compare p-value with α	$0.0038 < 0.05 \rightarrow$ p-value is less than α .

Industry-oriented conclusion

Since the p-value (0.0038) is less than α (0.05), we reject H_0 . There is statistically significant evidence that the new appointment scheduling system has reduced the average patient waiting time below 42 minutes. The hospital can conclude, with reasonable confidence, that the new system is effective and consider rolling it out more broadly.

Question 18 — Social Media Analytics

A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with standard deviation 0.8 hours. Determine whether the data provide sufficient evidence to reject the company's claim at $\alpha = 0.05$.

Step-by-step solution

1. Population	All users of the social media platform.
2. Sample	100 randomly selected users.
3. Parameter of interest	Population mean daily usage time, μ .
4. Point estimate	$\bar{x} = 3.2$ hours (sample standard deviation $s = 0.8$ hours).
5. Null hypothesis H_0	$\mu = 3$ hours (the company's claim is correct).
6. Alternative hypothesis H_1	$\mu \neq 3$ hours — two-tailed test.
7. Significance level α	$\alpha = 0.05$.
8. Test statistic	$SE = s / \sqrt{n} = 0.8 / \sqrt{100} = 0.8 / 10 = 0.08$ $z = (\bar{x} - \mu_0) / SE = (3.2 - 3) / 0.08 = 2.50$

9. p-value	$\text{p-value} = 2 \times P(Z > 2.50) = 2 \times (1 - 0.9938) = 2 \times 0.0062 = 0.0124.$
10. Compare p-value with α	$0.0124 < 0.05 \rightarrow \text{p-value is less than } \alpha.$

Industry-oriented conclusion

Since the p-value (0.0124) is less than α (0.05), we reject H_0 . There is sufficient statistical evidence to reject the company's claim that average daily usage is exactly 3 hours; the true average usage time appears to be higher than claimed. The company should update its reported usage statistics and may use this finding to attract advertisers with a more accurate (higher) engagement figure.

Question 19 — Machine Learning

A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations. Formulate suitable null and alternative hypotheses and determine whether the improvement is statistically significant using a p-value.

Step-by-step solution

1. Population	All possible predictions the classification model could make on the underlying data distribution.
2. Sample	500 randomly selected test observations.
3. Parameter of interest	Population proportion of correctly classified observations (accuracy), p .
4. Point estimate	$\hat{p} = 0.86$ (86%).
5. Null hypothesis H_0	$p = 0.82$ (the new model's accuracy is the same as the existing model).
6. Alternative hypothesis H_1	$p > 0.82$ (the new model's accuracy is higher) — right-tailed test.
7. Significance level α	$\alpha = 0.05$.
8. Test statistic	$SE = \sqrt{[p_0(1 - p_0) / n]} = \sqrt{[(0.82 \times 0.18) / 500]} = \sqrt{0.0002952} = 0.01718$ $z = (\hat{p} - p_0) / SE = (0.86 - 0.82) / 0.01718 = 2.33$

9. p-value	$\text{p-value} = P(Z > 2.33) = 1 - 0.9901 = 0.0099.$
10. Compare p-value with α	$0.0099 < 0.05 \rightarrow \text{p-value is less than } \alpha.$

Industry-oriented conclusion

Since the p-value (0.0099) is less than α (0.05), we reject H_0 . The 4-percentage-point improvement in accuracy (from 82% to 86%) is statistically significant. The data science team can conclude that the new classification model genuinely outperforms the existing model and is justified in deploying it to production.

Question 20 — Manufacturing / Quality Control

A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective. Perform a hypothesis test at the 5% significance level using both (a) confidence interval and (b) p-value approaches. Compare the conclusions.

Step-by-step solution

1. Population	All products manufactured by the production process.
2. Sample	2,000 randomly selected products.
3. Parameter of interest	Population proportion of defective products, p .
4. Point estimate	$\hat{p} = 55 / 2000 = 0.0275$ (2.75%).
5. Null hypothesis H_0	$p = 0.02$ (the true defect rate is 2%, as claimed).
6. Alternative hypothesis H_1	$p \neq 0.02$ (the true defect rate differs from 2%) — two-tailed test.
7. Significance level α	$\alpha = 0.05$.
8a. Confidence interval (using $\hat{p}$)	$SE(CI) = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[(0.0275 \times 0.9725) / 2000]} = 0.003657$ $95\% \text{ CI} = \hat{p} \pm 1.96 \times SE = 0.0275 \pm 0.00717 = (0.0203, 0.0347)$

	H_0 value $p_0 = 0.02$ lies OUTSIDE this interval.
8b. Test statistic (using p_0)	$SE(\text{test}) = \sqrt{[p_0(1 - p_0) / n]} = \sqrt{[(0.02 \times 0.98) / 2000]} = 0.003131$ $z = (\hat{p} - p_0) / SE(\text{test}) = (0.0275 - 0.02) / 0.003131 = 2.40$
9. p-value	$p\text{-value} = 2 \times P(Z > 2.40) = 2 \times (1 - 0.9918) = 2 \times 0.0082 = 0.0164.$
10. Compare	CI approach: 0.02 falls outside the 95% CI (0.0203, 0.0347) $\rightarrow$ reject H_0 . p-value approach: $0.0164 < 0.05 \rightarrow$ reject H_0 .

Industry-oriented conclusion

Both approaches agree. The 95% confidence interval for the true defect rate, (2.03%, 3.47%), does not contain the claimed value of 2%, and the p-value (0.0164) is less than α (0.05). We therefore reject H_0 and conclude that the true defect rate is significantly higher than the manufacturer's claim of 2%. The company should investigate the production process to identify the cause of the higher-than-claimed defect rate before it affects customer satisfaction and warranty costs.